Game-Theoretic Cosmology and the Curvature Relaxation Model: A Nash Equilibrium Framework for Cosmic Acceleration

L. Geiger[1, *]

[1] *Independent Researcher, Bernau, Germany*[†]

(Dated: June 5, 2026)

We develop a game-theoretic framework for cosmology in which spacetime evolution is modeled as a Nash equilibrium between a metastable quantum vacuum (null space) and a spacetime bubble. The central result is the *Curvature Relaxation Model* (CRM), which explains accelerated expansion not through dark energy but through a diminishing curvature return potential $\Omega_\Phi(a) = \Phi_0 \cdot \tanh(k(a - a_{\text{trans}}))$ – a geometric "memory" of the Big Bang energy concentration. Tested against 1,590 Pantheon+ SN Ia data points [20] with both diagonal and full covariance errors, the CRM yields $\Delta\chi^2 = -12.2$ (-11.2) and $\Delta\text{AIC} = -8.2$ (-7.2) relative to ΛCDM; the 5-fold cross-validation mean is slightly lower for CRM but, without fold-level variance estimates, should be treated as comparable rather than decisive. MCMC analysis gives $\Omega_m = 0.368 \pm 0.024$; four alternative saturation functions yield comparable results, confirming robustness. The phenomenological Pantheon+ fit produces a phantom-like effective diagnostic, $w_{\text{eff}}(z) < -1$, but current DESI DR2 constraints favor $w_0 > -1$, $w_a < 0$ in the relevant $w_0 w_a$CDM combinations [19]; persistent phantomness is therefore treated as a tension to be resolved, not as a fundamental CRM prediction. The more robust late-time signature is a later acceleration onset ($z_{\text{acc}} = 0.52$ vs. 0.84), testable with Euclid, Roman, and DESI. The later acceleration implies an extended matter-dominated growth phase, naturally explaining the JWST "Universe Breakers" at $z > 7$ [23] and the El Gordo anomaly [25]. Phantom pathologies are absent in the fitted ansatz because Ω_Φ saturates ($\Omega_\Phi \to \Phi_0$, de Sitter end state, no Big Rip). The game-theoretic axioms possess an exact thermodynamic equivalence connecting to the Jacobson tradition [31]. Analysis code is publicly available.[a]

I. INTRODUCTION

The discovery of the accelerated expansion of the universe through observations of distant Type Ia supernovae in 1998 by the teams of Perlmutter [2] and Riess and Schmidt [1] marks a turning point in modern cosmology. The 2011 Nobel Prize in Physics was awarded for this discovery. The standard model of cosmology, ΛCDM, explains the acceleration through a cosmological constant Λ comprising approximately 68% of the energy density of the universe [3]. Despite its empirical success, ΛCDM faces profound conceptual problems:

1. **The Cosmological Constant Problem:** The observed vacuum energy density is ∼60–120 orders of magnitude smaller than theoretical predictions from quantum field theory [4].

2. **The Coincidence Problem:** Why are Ω_m and Ω_Λ of comparable magnitude precisely in the present epoch?

3. **The H_0 Tension:** The local measurement of the Hubble parameter ($H_0 \approx 73\,\text{km/s/Mpc}$) differs significantly from the CMB-derived value ($H_0 \approx 67.4\,\text{km/s/Mpc}$) [3, 5].

Recent results from the *Dark Energy Spectroscopic Instrument* (DESI) strengthen doubts about a strictly constant dark energy: the DESI DR1 analysis of baryon acoustic oscillations combined with CMB and supernova data shows a 2.5–3.9σ preference for a model with time-dependent equation-of-state parameter $w(z)$ over ΛCDM [6]. The subsequent DESI DR2 result [19] reinforces this finding with higher statistical power: in $w_0 w_a$CDM it favors the quadrant $w_0 > -1$, $w_a < 0$, with supernova-dependent preferences over ΛCDM ranging from 2.8–4.2σ. For example, the DESI+CMB+Union3 combination gives $w_0 = -0.667 \pm 0.088$, $w_a = -1.09^{+0.31}_{-0.27}$. This is in potential tension with the phantom-like effective diagnostic of the phenomenological CRM fit ($w_{\text{eff}} < -1$); see the Limitations discussion (Sec. VIII) for a detailed assessment.

In parallel, theoretical works show that the acceleration could also be explained without dark energy: Pfeifer et al. [7] demonstrate within the framework of Finsler gravity that a generalized spacetime geometry naturally produces exponential expansion in vacuum. Trivedi and Venkatasubramanian [8] show in their *Cosmological Teledynamics* that the universe operates like a "giant potential game" converging toward a continuous Nash equilibrium, where cosmic acceleration emerges as an emergent effect of dynamic memory in a self-gravitating medium.

This paper connects these developments with an inde-

* https://orcid.org/0009-0005-7296-1534

† AI tools (Claude, Anthropic; Copilot, Microsoft/GitHub; Gemini, Google DeepMind) were used for mathematical formalization, code development, and text generation. All physical hypotheses, scientific interpretation, and responsibility for the content lie solely with the author. The analysis code is available at `https://github.com/research-line/crm-cosmology`.; Paper I in the CRM series.

a `https://github.com/research-line/crm-cosmology` pendent approach: starting from a game-theoretic model of the interaction between quantum vacuum and spacetime, the *Curvature Relaxation Model* (CRM) is developed, which interprets accelerated expansion as a "releasing brake" rather than a "new drive."

II. GAME-THEORETIC FRAMEWORK: NULL SPACE AND SPACETIME BUBBLE

Epistemological status. The game-theoretic framework developed in this section is a *heuristic-conceptual* interpretive layer in the tradition of Jacobson's thermodynamic derivation of the Einstein equations [31]. It provides physical intuition and motivates the functional form of the CRM, but it is not the operative dynamical framework. The operative dynamics is the Lagrangian formulation $\mathcal{L} = R/(16\pi G) + \gamma R^2 + \frac{1}{2}(\partial\phi)^2 - V_{\mathrm{PT}}(\phi)$ derived in Paper III [42], from which all testable predictions follow. The game-theoretic picture should therefore be understood as an *interpretive meta-framework*, not as an alternative fundamental dynamics.

A. Fundamental Assumptions

The proposed framework rests on the following assumptions:

1. There exists a metastable quantum vacuum state (hereafter: *null space*) characterized by quantum fluctuations, with observable vacuum effects such as the Casimir force as the minimal physical analogy [16]. *Ontological caveat:* The null space is a *pre-geometric abstraction* – a boundary condition for the variational problem – not a concrete physical entity with independent spatiotemporal existence. Multiple UV-completion programs (Loop Quantum Gravity, Causal Set Theory, quantum error correction; see Paper III [42], Appendix) are compatible with this abstraction.

2. An extraordinarily large fluctuation extracts a one-time energy amount E_0 from the null space, creating a concentration gradient.

3. To encapsulate and controllably neutralize this gradient, spacetime emerges as a dynamic structure – the *spacetime bubble* (daughter system).

4. A game-theoretic equilibrium exists between the null space (parent system) and the spacetime bubble.

These assumptions are formalized in the following sections.

B. Agents and Objectives

The system is modeled as a two-player potential game:

Null Space (Parent System):: The primary objective is self-protection – preservation of its structural integrity. It regulates the coupling strength to the spacetime bubble via effective boundary conditions ("gatekeeping"), enforces slow energy dissipation (damping), and forms buffer zones (horizon-like shells).

Spacetime Bubble (Daughter System):: The primary objective is the controlled return to the null state while simultaneously protecting the parent system. Strategies include cascaded gradient reduction, adiabatic return, and entropy management.

C. Mathematical Formulation as a Potential Game

The global objective function of the system reads:

$$\Phi = \alpha \cdot S_{\mathrm{parent}} + \beta \cdot R_{\mathrm{daughter}} - \gamma \cdot G \tag{1}$$

where S_{parent} denotes the structural integrity of the null space, R_{daughter} the return progress, and G the remaining concentration gradient; $\alpha, \beta, \gamma > 0$.

Definition 1 (Nash Equilibrium of the Cosmological Game)**.** *A strategy pair (s_P^*, s_D^*) of null space and spacetime bubble constitutes a Nash equilibrium if:*

$$\Phi(s_P^*, s_D^*) \geq \Phi(s_P, s_D^*) \quad \forall\, s_P \tag{2}$$
$$\Phi(s_P^*, s_D^*) \geq \Phi(s_P^*, s_D) \quad \forall\, s_D \tag{3}$$

Neither side can improve the overall potential by unilaterally deviating from its strategy without endangering the stability of the system.

This is the standard unilateral-deviation criterion introduced by Nash [18], used here only as an interpretive two-player potential-game template.

The central **conflict** is that excessively rapid reduction of G (immediate return) endangers S_{parent}, while excessively slow reduction increases entropy and costs within the bubble. The Nash equilibrium therefore enforces a controlled, temporally extended neutralization.

D. Emergent Laws from the Equilibrium

Caveat on the game-theoretic framework: The game-theoretic construction presented in Secs. II A–II E provides *interpretive motivation* for the CRM phenomenology—it does not constitute a formal derivation of the CRM potential $\Omega_\Phi(a)$ from first principles. The functional form $\Omega_\Phi(a) \propto \tanh(k(a - a_{\mathrm{trans}}))$ was chosen to fit Type Ia supernova data; the game-theoretic narrative was constructed to make this choice physically intuitive. This is a legitimate methodological approach in phenomenological cosmology, but readers should be aware that the game-theoretic argument does not uniquely determine the CRM functional form. A formal derivation from a quantum gravity or effective field theory action remains outstanding.

From the game-theoretic equilibrium condition, physical regularities emerge (as heuristic consequences of the Nash equilibrium picture):

1. **Energy conservation:** Conservative field equations arise as a necessary condition for stable gradient reduction.

2. **Causal structure:** Shell formation by the null space enforces a maximum propagation speed for information and energy.

3. **Entropic arrow of time:** "Time" within the bubble is the order along which the concentration gradient is leveled.

4. **Flux limitation:** Maximum fluxes across the shell scale sublinearly with the internal excess, preventing runaway processes.

5. **Asymptotic return:** The residual gradient $G \to 0$ approaches zero only asymptotically; there is no catastrophic finale.

The last property is particularly significant for cosmology: it implies that the expansion of the universe never reverses but proceeds asymptotically – consistent with observational data.

E. Thermodynamic Equivalence

The game-theoretic framework of the preceding sections has a precise equivalence in the language of thermodynamics and variational physics. This connection reflects a deep mathematical isomorphism between game theory and statistical mechanics [34]:

- **Nash equilibrium ↔ Thermodynamic equilibrium:** The state from which no subsystem can improve its objective unilaterally corresponds to the state of maximum entropy under constraints.

- **Potential function Φ ↔ Entropy functional:** The global quantity extremized by the equilibrium strategies is the entropy of the coupled system.

- **Optimal strategy selection ↔ Principle of least action:** Each agent's strategy corresponds to field equations arising from $\delta S = 0$.

- **Gradient reduction rate ↔ Maximum Entropy Production Principle (MEPP):** Among all paths consistent with the boundary conditions, the system evolves along the one that maximizes entropy production [33].

This equivalence is grounded in black-hole thermodynamics [17] and in a foundational result: Jacobson [31] showed that the Einstein field equations can be derived from the proportionality of entropy and horizon area together with the Clausius relation $\delta Q = T\,dS$, establishing that general relativity itself is a thermodynamic equation of state. Recent work has extended this paradigm to non-Riemannian geometries incorporating torsion and non-metricity [32], suggesting that the thermodynamic route to gravity is more general than the metric framework and may naturally accommodate the additional geometric degrees of freedom (such as the CRM scalaron) as entropy-maximizing channels. If standard gravity follows from thermodynamics, it is natural to expect that *modified* gravity – here, the curvature relaxation mechanism – follows from modified thermodynamic boundary conditions, namely the constraint that the parent vacuum (null space) remains structurally stable while entropy production in the daughter system is maximized.

The saturation mechanism of Section III C is thus directly interpretable as an approach to thermodynamic equilibrium: the tanh dynamics describes the entropy production rate of a system with finite capacity that evolves under the joint constraint of the second law and parent-system stability.

III. THE CURVATURE RELAXATION MODEL (CRM)

A. Physical Motivation

In the game-theoretic framework of the preceding section, spacetime is interpreted as a "braking mechanism" that prevents the immediate return of energy to the null space. The central physical insight is:

> *The observed accelerated expansion is not caused by a new form of energy but by a diminishing curvature return potential – a kind of geometric "memory" of the initial energy concentration at the Big Bang.*

The analogy is that of a stretched spring: initially, maximum tension (high curvature) produces strong restoring force. Over time, the tension relaxes, the restoring force decreases, and the expansion "accelerates" relative to the braked early phase – like a car whose handbrake is slowly released.

B. Modified Friedmann Equation

The standard Friedmann equation in the ΛCDM model reads:

$$H^2(a) = H_0^2 \left[\Omega_m\, a^{-3} + \Omega_\Lambda\right] \tag{4}$$

In the CRM, the cosmological constant is replaced by a time-dependent curvature return potential:

$$H^2(a) = H_0^2 \left[\Omega_m\, a^{-3} + \Omega_\Phi(a)\right] \tag{5}$$

The curvature return potential is defined as:

$$\Omega_\Phi(a) = \Phi_0 \cdot \frac{\tanh\big(k \cdot (a - a_{\text{trans}})\big) + s}{1 + s} \tag{6}$$

where $s = \tanh(k \cdot a_{\text{trans}})$ is a normalization shift ensuring $\Omega_\Phi(0) = 0$, and:

- a is the scale factor ($a = 1$ today, $a \to 0$ at the Big Bang),
- Φ_0 is the amplitude (derived from the flatness constraint $\Omega_m + \Omega_\Phi(1) = 1$),
- k is the transition sharpness,
- a_{trans} is the transition scale factor.

The specific parameter values are determined from the fit to the Pantheon+ data set in Section IV.

C. Dynamic Saturation Mechanism

The tanh parameterization is not an *ad hoc* chosen fitting function but arises as the exact solution of a physically motivated **dynamic saturation mechanism**. The central assumption is: the spacetime bubble possesses a finite absorption capacity for the curvature return. The return rate is proportional to the remaining capacity:

$$\frac{d\Omega_\Phi}{da} = k \cdot \left[1 - \left(\frac{\Omega_\Phi}{\Phi_0}\right)^2\right] \tag{7}$$

This equation describes a classical saturation process from dynamical systems theory: at small Ω_Φ, the potential grows nearly linearly (the "brake" releases at full rate); at $\Omega_\Phi \to \Phi_0$, saturation occurs (maximum capacity is reached, the brake is fully released). The exact solution of Eq. (7) is:

$$\Omega_\Phi(a) = \Phi_0 \cdot \tanh\big(k \cdot (a - a_{\text{trans}})\big) \tag{8}$$

where a_{trans} represents the integration constant (transition point). The saturation mechanism is ubiquitous in physics and appears in formally identical form in numerous systems:

- Ferromagnetism: spontaneous magnetization $M(T) \sim \tanh(T_C/T)$
- BCS superconductivity: energy gap $\Delta(T) \sim \tanh(T_C/T)$
- Soliton physics: kink solution $\phi(x) = \phi_0 \tanh(kx)$
- Nonlinear optics: saturation absorption $\alpha(I) \propto 1/(1 + I/I_{\text{sat}})$

All these systems share the property of an ordered transition from one state to another with finite capacity – *exactly* the behavior that the game-theoretic framework predicts for the Nash equilibrium between null space and spacetime bubble. The tanh form is thus not postulated but *derived* from the underlying mechanism. *Microscopic realization:* The saturation ODE (7) is not merely postulated at the phenomenological level but is later shown to emerge microscopically from a Pöschl-Teller potential in the effective Lagrangian (Paper III [42]): the $\cosh^{-2}$ potential admits tanh-type kink solutions as universal mean-field dynamics, providing a rigorous field-theoretic foundation for the saturation mechanism.

To verify robustness, four different saturation functions were tested (Section IV G). All yield $\Delta\chi^2 \approx -9$ to -12 relative to ΛCDM – the data "see" a saturation process, independent of the exact mathematical formulation.

D. Physical Interpretation of the Parameters

Early times ($a \to 0$, $z \to \infty$): $\Omega_\Phi \to 0$. The "brake" operates at full strength – the expansion follows matter dominance as in ΛCDM. There is no dark component.

Transition epoch ($a \approx a_{\text{trans}}$, $z \approx 1.5$): Ω_Φ increases. The "brake" begins to release. This occurred approximately 10.3 billion years ago.

Today ($a = 1$, $z = 0$): $\Omega_\Phi \to \Phi_0$. The maximum effect is reached; the potential effectively acts like Λ.

E. Effective Equation-of-State Parameter

The effective equation-of-state parameter of the curvature return potential is:

$$w_{\text{eff}}(a) = -1 - \frac{1}{3}\frac{d\ln\Omega_\Phi}{d\ln a} \tag{9}$$

Its time evolution is presented in Table I.

Within the phenomenological Pantheon+ tanh fit, the effective diagnostic in Table I is consistently below $w = -1$. This should not be read as a fundamental phantom-field prediction: it is a diagnostic of the fitted ansatz and must be confronted with the DESI DR2 preference for $w_0 > -1$, $w_a < 0$ and with the Lagrangian formulation of Paper III. The relevant falsification test is therefore whether the curvature-relaxation sector can reproduce the late-time expansion history and the later acceleration

TABLE I. Time evolution of the effective equation-of-state parameter $w_{\text{eff}}(z)$: ΛCDM vs. CRM. The 1σ uncertainties are from the MCMC posterior analysis (Section IV F).

z	w (ΛCDM)	w (CRM)	1σ range	Δw
0.0	−1.000	−1.355	[−1.371; −1.339]	**−0.355**
0.3	−1.000	−1.433	[−1.645; −1.355]	**−0.433**
0.5	−1.000	−1.450	[−1.730; −1.358]	**−0.450**
0.8	−1.000	−1.456	[−1.759; −1.359]	**−0.456**
1.0	−1.000	−1.454	[−1.749; −1.359]	**−0.454**
1.5	−1.000	−1.444	[−1.696; −1.357]	**−0.444**
2.0	−1.000	−1.432	[−1.644; −1.355]	**−0.432**

transition without requiring a persistent physical phantom component.

IV. NUMERICAL TESTS AND MODEL COMPARISON

A. Flatness Constraint

To reduce the number of free parameters and ensure physical consistency, the flatness constraint

$$\Omega_m + \Omega_\Phi(a{=}1) = 1 \tag{10}$$

is imposed. This yields for the amplitude:

$$\Phi_0 = \frac{(1-\Omega_m)(1+s)}{\tanh\big(k\cdot(1-a_{\text{trans}})\big)+s} \tag{11}$$

The CRM thus has three cosmological degrees of freedom (Ω_m, k, a_{trans}) plus one nuisance parameter (M), totaling four effective parameters – only two more than ΛCDM.

B. Data: Pantheon+

The test is performed against the Pantheon+ data set [20], a large public catalog of spectroscopically confirmed Type Ia supernovae. The release contains 1,701 light curves for 1,550 unique SNe Ia; after applying the $z > 0.01$ mask used here to avoid peculiar-velocity dominance, 1,590 SN Ia light-curve data points enter the fit, spanning $z = 0.0102$ to $z = 2.2614$. The observable is the bias-corrected apparent B-band magnitude `m_b_corr`. The analysis is performed both with diagonal errors and with the full statistical-systematic covariance matrix (STAT+SYS) of the Pantheon+ data set.

C. Methodology

Distance computation: The luminosity distance is computed via cumulative trapezoidal integration on a fine z-grid ($N = 2{,}000$ support points) and interpolated onto the data redshifts. This procedure is numerically stable (error $< 10^{-5}$) and enables fast evaluation during optimization.

Nuisance parameter: The absolute magnitude offset $M = M_B + 5\log_{10}(c/H_0) + 25$, which absorbs the absolute magnitude and the Hubble constant, is analytically marginalized:

$$M_{\text{best}} = \frac{\sum_i w_i(m_i^{\text{obs}} - \mu_i^{\text{th}})}{\sum_i w_i}, \quad w_i = \sigma_i^{-2} \tag{12}$$

Optimization: Parameter determination via *Differential Evolution* (global evolutionary optimizer) with subsequent L-BFGS-B refinement (*polish*).

MCMC uncertainties: For the flat CRM, parameter uncertainties are determined using *emcee* [22] (32 walkers, 3,000 steps, 500 burn-in). The posterior distributions yield 1σ confidence intervals for all parameters including derived quantities Φ_0 and z_{trans}.

Model selection: In addition to χ^2, the Akaike Information Criterion (AIC $= \chi^2 + 2k$) and the Bayesian Information Criterion (BIC $= \chi^2 + k\ln n$) are computed, where k is the number of effective parameters and n the number of data points. To check for overfitting, a 5-fold cross-validation is additionally performed. The complete analysis code is publicly available.[1]

D. Results

Three models are fitted: flat ΛCDM (2 parameters), CRM with flatness constraint (4 parameters), and CRM without constraint (5 parameters).

TABLE II. Fitted parameters and goodness of fit: ΛCDM vs. CRM against Pantheon+ (1,590 SN Ia data points after the $z > 0.01$ mask). For CRM (flat), 1σ MCMC uncertainties are given.

	ΛCDM	CRM (flat)	CRM (free)
Free parameters k	2	4	5
Ω_m	0.244	$0.368^{+0.025}_{-0.023}$	0.552
Ω_Λ / $\Omega_\Phi(z{=}0)$	0.756	0.636	0.872
Φ_0 (derived)	–	$0.988^{+0.615}_{-0.221}$	1.292
k (transition sharpness)	–	$1.44^{+1.22}_{-0.84}$	1.98
a_{trans} (z_{trans})	–	0.75 (0.33)	0.80 (0.25)
Ω_{total}	1.000	1.000	1.423
χ^2 (diagonal)	729.0	716.8	715.9
χ^2 (full cov.)	1432.0	1420.8	–
χ^2/dof	0.459	0.452	0.452
AIC (diagonal)	733.0	724.8	725.9
AIC (full cov.)	1436.0	1428.8	–
BIC	743.7	746.3	752.8

The CRM with flatness constraint yields $\Omega_m = 0.368 \pm 0.024$ (MCMC). *Tension note:* This value exceeds the

[1] https://github.com/research-line/crm-cosmology

Planck CMB value $\Omega_m^{\text{Planck}} = 0.315 \pm 0.007$ by $\Delta\Omega_m = 0.053$, corresponding to $\approx 2.1\sigma$ (adding the uncertainties in quadrature: $\sigma_{\text{tot}} = \sqrt{0.024^2 + 0.007^2} \approx 0.025$). This discrepancy is not a minor offset; it may indicate that the phenomenological $\Omega_\Phi(a)$ potential partially absorbs matter energy density, or that the SN Ia-only fit is insufficient to constrain Ω_m without CMB and BAO data. A joint fit including CMB and BAO is required to assess this tension. The fitted transition redshift $z_{\text{trans}} = 0.33$ ($a_{\text{trans}} = 0.75$) lies at later cosmic times than theoretically expected. The transition sharpness $k = 1.44^{+1.22}_{-0.84}$ describes a smooth transition with a broad posterior – the data prefer a transition but allow a range of transition sharpnesses.

Note on χ^2/dof < 1: The reduced chi-squared values (χ^2/dof $= 0.452$–0.459) are substantially below 1.0 for both models. This is characteristic of Pantheon+ SN Ia analyses using the diagonal covariance matrix, which is known to overestimate individual measurement uncertainties (statistical scatter exceeds the systematic floor). The effect is not model-specific: both ΛCDM and CRM show the same χ^2/dof ≈ 0.46, confirming that the AIC/BIC differences are not driven by error overestimation but reflect genuine differences in fit quality. The full covariance matrix analysis (χ^2/dof ≈ 0.91, approaching unity) provides a more calibrated error model.

Full covariance matrix: Repeating the analysis with the full statistical-systematic covariance matrix confirms the results: $\Delta\chi^2 = -11.2$ and ΔAIC $= -7.2$ (compared to -12.2 and -8.2 with diagonal errors). The slight reduction is explained by the inclusion of systematic correlations between neighboring supernovae.

E. Model Selection

Interpretation: Two selection criteria favor the flat CRM over ΛCDM outright, χ^2 (-12.2) and AIC (-8.2), while the raw cross-validation mean is slightly lower for CRM (0.4499 vs. 0.4519). The BIC, which more strongly penalizes additional parameters, favors ΛCDM (ΔBIC $= +2.6$). According to the Kass–Raftery scale [21], this lies in the "positive evidence" range against the more complex model. Cross-validation is the most direct overfitting check, but no fold-level standard deviation is reported; the 0.4% difference should therefore be treated as comparable generalization, not as decisive evidence for CRM.

F. MCMC Posterior Analysis

Parameter uncertainties are determined using *emcee* [22] (32 walkers, 3,000 steps, 500 burn-in, acceptance rate: 63%). *Convergence note:* Formal convergence diagnostics (Gelman–Rubin $\hat{R}$ statistic [38]; effective sample size N_{eff}) were not reported in the original analysis. The acceptance rate of 63% lies within the standard emcee target range (20–70% %), and visual inspection of the walker trajectories showed no obvious non-convergence, but quantitative convergence certification (e.g., $\hat{R} < 1.01$, N_{eff}/chain > 50) is recommended for publication-ready results. The results for the flat CRM are given in Table II as 1σ uncertainties. The posterior distribution of Ω_m is nearly Gaussian with $\Omega_m = 0.368^{+0.025}_{-0.023}$. The transition sharpness k shows a broad, asymmetric posterior ($k = 1.44^{+1.22}_{-0.84}$), meaning the data prefer a transition but constrain the sharpness less strongly. The derived quantities $\Phi_0 = 0.988^{+0.615}_{-0.221}$ and $z_{\text{trans}} = 0.35$ (MCMC posterior median; the point estimate in Table II gives 0.33) are consistent within the broad posterior of a_{trans}. The MCMC-computed $w(z)$ confidence bands (Table I) show that $w = -1$ lies outside the 1σ range for all redshifts.

Interpretation of $\Omega_m = 0.368$: The CRM value exceeds the Planck CMB value ($\Omega_m^{\text{Planck}} = 0.315 \pm 0.007$), which is typical for pure supernova fits. In the CRM context, this deviation has a possible physical interpretation: the model may be absorbing part of the density classified as "dark energy" in ΛCDM as a dynamic curvature effect. Since $\Omega_\Phi(a)$ vanishes at early times (unlike Ω_Λ = const.), Ω_m must be compensatorily higher to maintain the overall fit. This should not be read as independent support from weak-lensing surveys: lensing constraints primarily probe combinations such as S_8 and have their own tensions with Planck, so a joint SN+CMB+BAO+lensing fit is required before the higher SN-only Ω_m can be interpreted physically.

G. Robustness: Alternative Functional Forms

To verify whether the results depend on the specific choice of the tanh form, four different saturation functions are tested under identical conditions:

Result: All four functional forms yield $\Delta\chi^2 \approx -9$ to -12 relative to ΛCDM. The tanh form is neither the only possible nor the "best-fitting" – it represents a robust class of saturation functions. This refutes the objection that the CRM results depend on a specific functional choice. The Ω_m values converge at ≈ 0.36–0.37 for all forms, underscoring the physical consistency.

H. Phantom Stability Analysis

The fitted equation-of-state parameter $w < -1$ (phantom regime) raises the legitimate question of stability. In ordinary phantom scalar field models, $w < -1$ leads to a growing energy density ($\rho \propto a^{-3(1+w)} \to \infty$ for $w < -1$, $a \to \infty$), which inevitably produces the "Big Rip" in finite time – the destruction of all bound structures in the universe [9]. **The CRM fundamentally circumvents this pathology** for three reasons:

1. **Saturation instead of divergence:** The dy-

TABLE III. Model comparison: CRM vs. ΛCDM. Negative values favor the CRM.

Criterion	CRM (flat) vs. ΛCDM	CRM (free) vs. ΛCDM
$\Delta\chi^2$	$\mathbf{-12.2}$	-13.1
ΔAIC	$\mathbf{-8.2}$	-7.1
ΔBIC	$+2.6$	$+9.0$
5-fold $\langle\chi^2/n\rangle$	**0.4499**	0.4498
ΛCDM: $\langle\chi^2/n\rangle$	0.4519	

TABLE IV. Comparison of alternative functional forms for $\Omega_\Phi(a)$. All models use the flatness constraint and have $k = 4$ parameters.

Functional form	χ^2	AIC	$\Delta\chi^2$ vs. ΛCDM	Ω_m
tanh (standard CRM)	716.8	724.8	-12.2	0.364
Logistic function	717.5	725.5	-11.5	0.368
Error function (erf)	716.7	724.7	-12.3	0.367
Power law	720.1	728.1	-8.9	0.364

namic saturation mechanism (Eq. 7) guarantees $\Omega_\Phi(a) \to \Phi_0$ for $a \to \infty$. The effective energy density remains *finite and bounded for all times.* This is the decisive difference from phantom scalar fields: whereas there ρ diverges, Ω_Φ saturates at its maximum value Φ_0.

2. **De Sitter end state:** $w_{\mathrm{eff}} \to -1$ for $a \to \infty$. The universe asymptotically approaches *exactly the same end state as ΛCDM* – a stable de Sitter space. The phantom regime is a transient phenomenon, not an end state.

3. **No Big Rip:** Since Ω_Φ saturates, neither the energy density nor the scale factor diverges in finite time. The Big Rip is excluded – not by an additional assumption but as a *direct consequence* of the saturation mechanism.

4. **Geometric rather than fluid interpretation:** The formal violation of the null energy condition ($\rho + p \geq 0$) is unproblematic because Ω_Φ represents *not a physical field* but a geometric property of spacetime. The energy conditions of general relativity apply to the energy-momentum tensor of physical fields, not to effective geometric terms. Analogous situations are well established in the literature: $f(R)$ gravity theories routinely exhibit effective $w < -1$ without generating physical instabilities [11].

In summary: the CRM exhibits "phantom behavior" without phantom pathologies. The universe does not end in a rip but in equilibrium.

I. Deceleration Parameter and H_0 Implications

Deceleration parameter $q(z)$: The deceleration parameter provides an additional, independently testable prediction. In the CRM, the transition from decelerated to accelerated expansion ($q = 0$) occurs at $z_{\mathrm{acc}} = 0.52$ – significantly later in cosmic time than in the ΛCDM model ($z_{\mathrm{acc}} = 0.84$). Furthermore, the CRM predicts a stronger present-day acceleration: $q_0^{\mathrm{CRM}} = -0.81$ compared to $q_0^{\Lambda\mathrm{CDM}} = -0.63$.

Implication for structure formation: A later onset of cosmic acceleration ($z_{\mathrm{acc}} = 0.52$ instead of 0.84) means that gravitationally bound structures (galaxy clusters, large-scale filaments) *could grow undisturbed for longer* before the acceleration suppressed growth. The matter-dominated era – during which gravity drives structure growth – lasted significantly longer in the CRM than in the standard model.

Empirical evidence for early massive structures: Several independent observations put the ΛCDM model under significant tension regarding structure formation:

1. **JWST "Universe Breakers":** The James Webb Space Telescope has discovered galaxies at $z > 7$ (approximately 500–700 Myr after the Big Bang) that are far more massive than permitted by hierarchical structure formation in ΛCDM [23]. Boylan-Kolchin [24] demonstrates quantitatively that the stellar mass density of these objects exceeds the available baryon budget within ΛCDM halos – a "timing problem" of structure formation.

2. **El Gordo (ACT-CL J0102−4915):** This extremely massive galaxy cluster at $z \approx 0.87$ with mass $M_{200} \approx 2.1 \times 10^{15}\, M_\odot$ represents a $> 6\sigma$ tension with ΛCDM, since an object of this mass at this age with the observed collision velocity is statistically nearly impossible [25].

3. **Protocluster SPT2349−56:** Already 1.4 Gyr after the Big Bang ($z = 4.3$), this protocluster exhibits at least 14 gas-rich galaxies with a total star formation rate of $\sim$6,500 $M_\odot$/yr – far more mature than predicted by ΛCDM [26].

The CRM offers a natural explanation for all three observations: since the acceleration only begins at $z_{\mathrm{acc}} = 0.52$ (instead of $z_{\mathrm{acc}} = 0.84$), gravitationally bound structures had more cosmic time for undisturbed growth. The

CRM prediction is directly testable through galaxy cluster counts and weak gravitational lensing surveys (Euclid, Vera C. Rubin Observatory).

TABLE V. Deceleration parameter $q(z)$: ΛCDM vs. CRM.

z	q (ΛCDM)	q (CRM)	Δq
0.0	-0.634	-0.805	-0.171
0.5	-0.217	-0.017	$+0.200$
1.0	$+0.082$	$+0.321$	$+0.240$
2.0	$+0.346$	$+0.467$	$+0.121$

H_0 **implications:** The nuisance parameter M absorbs both the absolute magnitude M_B and H_0 via the relation $M = M_B + 5\log_{10}(c/H_0) + 25$. Using the SH0ES calibration ($M_B = -19.253$), the CRM yields $H_0 = 76.1\,\mathrm{km/s/Mpc}$ compared to $H_0 = 75.5\,\mathrm{km/s/Mpc}$ in ΛCDM – a difference of $\Delta H_0 = +0.5\,\mathrm{km/s/Mpc}$. The H_0 tension is not resolved by the CRM alone, as the difference lies within the measurement uncertainty. A direct resolution requires the combination with CMB and BAO data.

V. COMPARISON WITH ALTERNATIVE MODELS

A. ΛCDM (Standard Model)

The ΛCDM model is extremely simple ($w = -1$, constant, two cosmological parameters) and fits all current data well. However, it suffers from the cosmological constant problem and the coincidence problem [4].

B. Quintessence

Quintessence models [9] postulate a dynamic scalar field ϕ with a time-dependent equation-of-state parameter. They can alleviate the coincidence problem but require a new field and its potential $V(\phi)$ with many free parameters.

C. Modified Gravity: $f(R)$ Theories

$f(R)$ gravity theories [10, 11] replace the Ricci scalar R in the Einstein–Hilbert action with a more general function. They offer a geometric explanation without dark energy but are mathematically complex and partly inconsistent with observations (gravitational lensing, CMB). The most widely studied late-time $f(R)$ model, Hu-Sawicki [30], satisfies solar system constraints through the chameleon mechanism while producing viable cosmic acceleration; it serves as an important benchmark for the Lagrangian analysis in Paper III [42].

D. Emergent Gravity (Verlinde)

Verlinde [12, 13] proposes that gravity is not a fundamental force but an emergent, entropic phenomenon. In de Sitter spaces, the entropy associated with the cosmological horizon leads to an additional "dark" gravitational force that could explain galaxy behavior without dark matter. In a closely related approach, Padmanabhan [14] derives cosmic expansion as an emergent phenomenon from the difference between surface and bulk degrees of freedom, with the cosmological constant arising from spacetime's approach to holographic equipartition – a perspective that shares structural elements with the CRM's equilibrium framework.

E. Finsler Gravity

Pfeifer et al. [7] extend general relativity through Finsler geometry, in which the metric depends not only on position but also on velocity:

$$g_{\mu\nu}(x,y) = \frac{1}{2}\frac{\partial^2 F^2}{\partial y^\mu \partial y^\nu}, \quad y = \frac{dx}{d\lambda} \tag{13}$$

The resulting Finsler–Friedmann equation produces exponential expansion even in vacuum – without a cosmological constant.

F. Cosmological Teleodynamics

Trivedi and Venkatasubramanian [8] formulate a game-theoretic cosmology that exhibits remarkable parallels to the approach presented here. Their *Cosmological Teleodynamics* describes the universe as a "giant potential game" converging toward a continuous Nash equilibrium. Cosmic acceleration appears as a "statistically emergent effect of dynamic memory in a self-gravitating medium" – a formulation that conceptually corresponds to the "geometric memory" of the CRM.

G. Synoptic Comparison

VI. COMPLEMENTARITY AND POSSIBLE UNIFICATION

A. Three Models, One Insight

All three approaches – CRM, Finsler gravity, and Cosmological Teleodynamics – share a fundamental insight:

> *"The accelerated expansion is not a new 'thing' but a property of the geometry or the statistical structure of the universe itself."*

TABLE VI. Synoptic comparison of cosmological models without dark energy.

Property	CRM	Finsler	Teleodynamics
Theor. basis	Standard GR + potential	Finsler geometry	Stat. mechanics + game theory
Mechanism	Releasing “brake”	Velocity-dep. metric	Dynamic memory
Dark energy	Not needed	Not needed	Not needed
Empirical test	Pantheon+ (1,590 SN data points, $\Delta\chi^2 = -12$)	Pending	Qualitative
Prediction	$w(z)$ time variation	Exp. expansion	Nash convergence
Complexity	Low (4 params.)	High	Medium

B. Hypothesis: CRM as an Effective Description

A fascinating possibility is that the three models describe different aspects of the same phenomenon. By analogy with the relationship between thermodynamics and statistical mechanics, the following hierarchy could hold:

- **Finsler gravity** (microscopic, fundamental): All moments of the 1-particle distribution function contribute to gravity.
- **CRM** (macroscopic, phenomenological): The time-dependent potential $\Phi(a)$ effectively encodes the contribution of higher moments.
- **Teleodynamics** (systemic, game-theoretic): The Nash equilibrium dynamics describes the global optimization.

Mathematically, this complementarity can be represented as a hierarchical relationship:

$$\underbrace{G_{\mu\nu}^{\text{Finsler}}(x,y)}_{\substack{\text{Finsler gravity}\\ \text{(microscopic)}}} \xrightarrow{\langle\cdot\rangle_{\text{eff}}} \underbrace{G_{\mu\nu} + 8\pi G\,\Omega_\Phi(a)\,g_{\mu\nu}}_{\substack{\text{CRM: modified}\\ \text{Einstein equation}}} \xleftarrow{\delta\Phi/\delta s_i = 0} \underbrace{\max_{s_P, s_D} \Phi(s_P, s_D)}_{\substack{\text{Teleodynamics:}\\ \text{Nash equilibrium}}} \tag{14}$$

where the left arrow describes the effective averaging over the velocity-dependent degrees of freedom of Finsler geometry and the right arrow represents the variational condition of the game-theoretic potential that determines the temporal behavior of $\Omega_\Phi(a)$.

VII. TESTABILITY AND PREDICTIONS

A. Observable Signatures

1. Equation-of-state diagnostic: The phenomenological Pantheon+ fit yields $|\Delta w| \approx 0.4$ relative to ΛCDM, but DESI DR2 moves the physical target away from persistent phantomness and toward a $w_0 > -1$, $w_a < 0$ trajectory. The ESA mission Euclid [15], the Nancy Grace Roman Space Telescope, and DESI can test whether this is better described by curvature relaxation or by a generic $w_0 w_a$CDM parametrization.

2. Deceleration parameter: The CRM predicts an earlier transition to accelerated expansion ($z_{\text{acc}} = 0.52$ vs. ΛCDM: $z_{\text{acc}} = 0.84$) as well as a stronger present-day acceleration ($q_0 = -0.81$ vs. -0.63). This is testable independently of $w(z)$.

3. Structure growth: A modified growth rate $f \cdot \sigma_8$ is predicted, measurable through weak gravitational lensing and galaxy cluster counts. Initial empirical hints are already provided by the JWST “Universe Breakers” at $z > 7$ [23], the El Gordo anomaly ($> 6\sigma$ tension with ΛCDM; [25]), and unexpectedly mature protoclusters at $z > 4$ [26] – all consistent with the CRM prediction of an extended growth phase.

4. CMB integral effects: A modified ISW effect (*Integrated Sachs–Wolfe*) in CMB temperature cross-correlations.

B. Future Missions

TABLE VII. Relevant observational missions for the CRM test.

Mission	Launch	$\sigma(w)$	Relevance
Euclid	2023	≈ 0.02	BAO + lensing
Roman	2026–2027	≈ 0.03	SN to $z \approx 2$
DESI	2021–	≈ 0.04	BAO + growth

C. Model Distinguishability

VIII. DISCUSSION

A. Strengths of the Approach

1. **Conceptual elegance:** No new form of energy required; the acceleration is a “releasing constraint,” not a “new drive.”

2. **Game-theoretic foundation:** The emergence of physical laws from equilibrium conditions offers a

TABLE VIII. Comparison of predictions: ΛCDM vs. CRM (Pantheon+ fit). The 1σ uncertainties are from the MCMC analysis.

Property	ΛCDM	CRM
$w(z{=}0)$	-1.000	-1.36 ± 0.02
$w(z{=}0.5)$	-1.000	$-1.45^{+0.09}_{-0.28}$
$w(z{=}1)$	-1.000	$-1.45^{+0.10}_{-0.30}$
$w(z{=}2)$	-1.000	$-1.43^{+0.08}_{-0.21}$
Time variation	None	Yes (consistently $w < -1$)
Δw (measurable)	–	≈ -0.4
q_0 (today)	-0.63	-0.81
$z_{\rm acc}$ (transition)	0.84	0.52

novel explanatory framework, independently supported by the work of Trivedi and Venkatasubramanian [8].

3. **Empirical validation:** The CRM fits 1,590 Pantheon+ SN Ia data points better than ΛCDM by χ^2 and AIC ($\Delta\chi^2 = -12.2$, $\Delta\text{AIC} = -8.2$) and shows comparable cross-validation performance (0.4499 vs. 0.4519, difference of 0.4%).

4. **Testability:** Specific, quantitative predictions for $w(z)$ and $z_{\rm acc}$ that are verifiable within a decade.

5. **Empirical support:** The CRM prediction of an extended growth phase ($z_{\rm acc} = 0.52$) offers a natural explanation for the JWST "early galaxy tension" [23, 24], the El Gordo anomaly [25], and unexpectedly mature protoclusters [26].

6. **Convergence of independent approaches:** CRM, Finsler gravity, and Cosmological Teleodynamics independently arrive at the same conclusion: dark energy is not necessary.

7. **Reproducibility:** Analysis code and data are publicly available (https://github.com/research-line/crm-cosmology).

B. Limitations and Open Questions

1. **Phenomenological character and tanh derivation:** The CRM is not a fundamental theory. The tanh functional form for $\Omega_\Phi(a)$ was chosen on phenomenological grounds (smooth saturation, physically motivated shape) and subsequently shown to be the exact solution of a saturation ODE (7). This logical order should be noted: the ODE was *constructed to have* tanh as a solution, not derived from first principles; the relationship is therefore motivational, not deductive. Four alternative functional forms yield comparable results (Section IV G), confirming that results are not specific to tanh, but also confirming that no particular shape is privileged by the current level of theory. A derivation from a fundamental quantum gravity or effective field theory action remains outstanding.

2. **Parameter freedom and BIC penalty:** Four effective parameters versus two in ΛCDM lead to a BIC disadvantage of $\Delta\text{BIC} = +2.6$. The BIC applies a stronger complexity penalty than the AIC ($k \ln n$ vs. $2k$), and a positive ΔBIC cannot be dismissed as marginal: on the Kass–Raftery (1995) scale, $\Delta\text{BIC} = 2$–6 constitutes "positive" evidence against the more complex model. The cross-validation performance (0.4499 vs. 0.4519, a difference of 0.4%) is consistent with the BIC verdict and does not provide decisive evidence *for* the CRM: without fold-level variance estimates, the 0.4% difference cannot be assessed for statistical significance (it could be within noise). An honest summary is: AIC favors CRM, BIC favors ΛCDM, cross-validation is inconclusive. Decisive model discrimination will require CMB and BAO data.

3. **Missing CPL benchmark:** The AIC advantage ($\Delta\text{AIC} = -8.2$) is computed only against ΛCDM (2 parameters). The w_0w_aCDM (CPL) model [39, 40] has the same parameter count as the CRM (flat) and also allows phantom crossing. Without a direct $\Delta\chi^2$ comparison against CPL, it cannot be excluded that the CRM's improved fit reflects a generic benefit of adding flexibility to the late-time equation of state rather than a specific signature of the curvature relaxation mechanism. Such a comparison is deferred to Paper II.

4. **Phantom regime:** The effective equation-of-state parameter $w < -1$ lies in the phantom regime. As shown in Section IV H, this leads in the CRM context neither to a Big Rip nor to instabilities, since Ω_Φ saturates and does not represent a physical field. Formally, the CRM violates the null energy condition, analogous to $f(R)$ gravity theories [11].

5. **Model comparison:** The comparison in this paper is restricted to ΛCDM. A fairer benchmark for the phantom crossing behavior would include the w_0w_aCDM (CPL) parametrization [39, 40], which also allows $w < -1$. Such a comparison is presented in Paper II.

6. **Outstanding tests:** CMB predictions (ISW effect, CMB power spectrum), BAO signatures, and gravitational lensing effects remain to be computed. The analysis with the full covariance matrix (Section IV D) does however confirm the results of the diagonal analysis.

7. **Microscopic basis:** What is Φ at the quantum level? The connection to a theory of quantum gravity is outstanding. The possible relationship to Finsler gravity (Section VI) could bridge this gap.

8. H_0 **tension:** The H_0 analysis (Section IV I) shows $\Delta H_0 = +0.5$ km/s/Mpc between CRM and ΛCDM – too small to resolve the H_0 tension. A resolution requires the combination with CMB and BAO data.

9. **DESI DR2 tension and resolution via the Lagrangian theory:** The phenomenological Pantheon+ fit in this paper yields $w_{\text{eff}} < -1$ (phantom-like diagnostic). However, DESI DR2 [19] reports preferred $w_0 w_a$CDM solutions in the quadrant $w_0 > -1$, $w_a < 0$; the published DESI+CMB+Union3 row gives $w_0 = -0.667 \pm 0.088$, $w_a = -1.09^{+0.31}_{-0.27}$, while Pantheon+ and DESY5 combinations give different but still non-phantom w_0 values. We now treat this as a direct tension with the original phenomenological tanh reading and defer a quantitative resolution to the Lagrangian formulation in Paper III [42]: the $f(R) = R + 2\gamma R^2$ sector is expected to approach $w = -1$ from above, rather than requiring a persistent physical phantom component. The value $w = -1.355$ is therefore a fitted effective diagnostic, not a prediction of the fundamental theory. Kovacs [44] is directionally relevant because its proposed covariant extension also lists $w \geq -1$ as a candidate consequence of the curvature-occupation kinetic structure, but that manuscript explicitly presents the cosmological extension as contingent and not yet a completed perturbation theory. We therefore regard $w \geq -1$ as the natural CRM target to test, not as a result already established by Paper I.

C. Philosophical Implications

If the CRM (or a related model) is confirmed, this would have profound consequences:

- **Dark energy is not a "thing":** It would be a geometric memory, not a physical field.
- **The universe "knows" about its beginning:** The geometry possesses a "memory."
- **Paradigm shift:** From "What drives the acceleration?" to "Why did the expansion brake earlier?"

This would be comparable to the transition from "What drives the planets?" (Ptolemy: spheres) to "How do planets move in the geometry of space?" (Kepler, Newton, Einstein).

D. Hierarchical Nash Equilibria

The two-player model of Section II (null space vs. spacetime bubble) represents only the first level of an emergent hierarchy. Paper IV [43] demonstrates that the Nash equilibrium of the cosmological game generates *daughter dynamics*: the scalar sector (scalaron) that provides the cosmological background is itself insufficient to stabilize dissipative structures (galaxies) at the galactic scale. A second equilibrium level – a timelike unit vector field A_μ, derived from the Maximum Entropy Production Principle – is required to produce flat rotation curves and the Radial Acceleration Relation. The hierarchical structure is thus:

Null space → *Scalar sector* (Paper III) → *Vector sector* (Paper IV) → *Standard physics.*

Each level is a Nash equilibrium that, when exhausted, necessitates the next degree of freedom. The two-player framework of this paper provides the conceptual template; Papers III and IV supply the Lagrangian realization.

E. Deductive Structure of the Model

A distinguishing feature of the CRM is that its Lagrangian is not assembled *ad hoc* from independent phenomenological terms, but follows from a deductive chain that begins with two assumptions and terminates at testable predictions:

1. **Axiom 1 (Initial condition):** A metastable quantum vacuum (null space) admits a large fluctuation that creates a concentration gradient – the nascent spacetime.

2. **Axiom 2 (Evolution principle):** The system evolves along the path that maximizes entropy production while preserving the structural integrity of the null space (equivalently: the principle of least action under the boundary condition of parent-system stability; see Section II E).

From these two assumptions, the following chain of deductions – presented here in retrospect; its rigorous derivation is the subject of Paper III – leads to the CRM action (Paper III, Eq. (8)):

- Axiom 2 requires a *controlled, time-extended* neutralization of the gradient → Einstein–Hilbert term $R/(16\pi G)$ (conservative field equations as necessary condition).
- The gradient has finite magnitude but the return must not destabilize the vacuum → *saturation* of the return rate → Pöschl-Teller scalar field with tanh-dynamics.
- Minimality (ghost-freedom + renormalizability at one loop) strongly favors R^2 as the leading gravitational correction (note: this represents a physically motivated selection principle, not a formal uniqueness theorem) → Starobinsky term γR^2.

- The scalaron (R^2 degree of freedom) produces an $a^{-\beta}$ power-law term that replaces particle dark matter at the cosmological level; trace coupling to $T_{\mu\nu}$ automatically suppresses this term during the radiation era (BBN protection).

- On galactic scales, the same R^2 scalaron enhances gravity by $\mu \to 4/3$ at sub-Compton scales (Paper III), coinciding with the MOND phase-space factor $V_3/V_2 = 4/3$ (Paper II).

- Chameleon screening ensures solar system compatibility without additional parameters.

The resulting Lagrangian $\mathcal{L} = R/(16\pi G) + \gamma R^2 + \frac{1}{2}(\partial\phi)^2 - V_{\rm PT}(\phi)$ has its *form* fully determined by the two axioms; only the *numerical values* of γ, V_0, and ϕ_0 are fixed by data. This deductive closure – from assumptions to a unique Lagrangian to falsifiable predictions – is what elevates the CRM from a phenomenological fit to a theoretical framework.

IX. CONCLUSION AND OUTLOOK

This paper has shown:

1. A game-theoretic framework for cosmology – the Nash equilibrium between null space and spacetime bubble – naturally leads to a model in which physical laws appear as emergent equilibrium conditions. The game-theoretic axioms possess an exact thermodynamic equivalence (Nash equilibrium $\leftrightarrow$ thermodynamic equilibrium, tanh saturation $\leftrightarrow$ MEPP dynamics), grounding the framework in the Jacobson tradition of gravity as a thermodynamic equation of state [31].

2. The resulting *Curvature Relaxation Model* (CRM) explains accelerated expansion without dark energy and passes the test against 1,590 Pantheon+ SN Ia data points after the $z > 0.01$ mask [20]: $\Delta\chi^2 = -12.2$ (diagonal) / -11.2 (full covariance matrix), ΔAIC $= -8.2$ / -7.2, and comparable cross-validation performance (0.4499 vs. 0.4519).

3. Model selection is mixed: χ^2 and AIC favor CRM, BIC favors ΛCDM, and 5-fold cross-validation is comparable but not decisive without fold-level variance estimates. Four alternative functional forms yield $\Delta\chi^2 \approx -9$ to -12, reducing but not eliminating the risk that the improvement reflects extra late-time flexibility.

4. MCMC-based parameter uncertainties ($\Omega_m = 0.368 \pm 0.024$) and the phantom stability analysis (no Big Rip, asymptotically de Sitter) support phenomenological consistency, while publication-ready convergence diagnostics and a direct CPL/CMB/BAO comparison remain necessary.

5. The CRM makes testable predictions: a later acceleration transition ($z_{\rm acc} = 0.52$ vs. 0.84 in ΛCDM) and a distinctive effective equation-of-state history that must be checked against DESI BAO, CMB, and supernova combinations rather than interpreted as persistent physical phantomness. Already now, the CRM prediction of an extended growth phase finds empirical support from JWST observations of unexpectedly massive galaxies at high redshifts [23, 24] and the statistically improbable existence of massive clusters such as El Gordo [25].

6. The convergence of three independent approaches (CRM, Finsler gravity, Cosmological Teleodynamics) points toward a possible paradigm shift: *dark energy as an independent entity may be superfluous.*

Next steps include: (a) testing against Planck CMB and DESI BAO data (the full Pantheon+ covariance matrix has already been taken into account), (b) computation of CMB power spectrum and structure growth predictions ($f\sigma_8$), (c) exploration of the connection between CRM and Finsler geometry, (d) development of a covariant formulation of $\Phi(a)$ from the Ricci scalar R, (e) investigation of quantum mechanical foundations of the curvature return potential, and (f) combination with local distance ladder data for direct H_0 determination.

Outlook: Unification with MOND – A universe without a dark sector? A particularly fascinating perspective opens up through the combination of the CRM with *Modified Newtonian Dynamics* (MOND) [27]. While the CRM eliminates dark energy as a geometric effect, MOND replaces dark matter through modified gravitational dynamics on galactic scales. Both frameworks converge on the prediction that structures form earlier and more efficiently than in ΛCDM – the CRM through an extended matter-dominated era ($z_{\rm acc} = 0.52$), MOND through effectively stronger gravity at low accelerations [25]. A preliminary analysis with a purely baryonic universe ($\Omega_m = \Omega_b \approx 0.05$) and an extended geometric potential $\Omega_\Phi(a) = \Phi_0 \cdot f_{\tanh}(a) + \alpha \cdot a^{-\beta}$ yields $\Delta\chi^2 = -26.3$ and ΔAIC $= -16.3$ relative to ΛCDM – *substantially outperforming both the standard CRM and ΛCDM.* The MCMC posterior analysis yields $\beta = 2.02 \pm 0.20$, which corresponds exactly to the scaling of spatial curvature (a^{-2}). "Dark matter" would thus be a dynamic curvature effect, not a particle. This result requires a full relativistic treatment (e.g., TeVeS or AeST [28, 29]) and will be analyzed in detail in a companion paper.

Software

This work uses `emcee` [22] for MCMC sampling, `NumPy` [35], `SciPy` [36] for numerical computations, and `Matplotlib` [37] for visualizations.

ACKNOWLEDGMENTS

The author thanks the developers of `CLASS` [41] and `emcee` [22] for making their codes publicly available.

a. AI tools. AI-based tools (Claude by Anthropic; Copilot by Microsoft/GitHub; Gemini by Google DeepMind) were used for code development, numerical analysis, literature review, and editorial support. All physical hypotheses, mathematical derivations, and scientific interpretation are solely the author's work.

b. Funding information. This research received no external funding.

c. Data availability. The Pantheon+ supernova data [20] are publicly available. All analysis code and numerical results are available at `https://github.com/research-line/crm-cosmology` (DOI: 10.5281/zenodo.18728936).

---

[1] Riess, A. G. et al. (1998). Observational Evidence from Supernovae for an Accelerating Universe and a Cosmological Constant. *The Astronomical Journal*, 116(3), 1009–1038. DOI: 10.1086/300499.

[2] Perlmutter, S. et al. (1999). Measurements of Ω and Λ from 42 High-Redshift Supernovae. *The Astrophysical Journal*, 517(2), 565–586. DOI: 10.1086/307221.

[3] Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 641, A6. DOI: 10.1051/0004-6361/201833910.

[4] Weinberg, S. (1989). The Cosmological Constant Problem. *Reviews of Modern Physics*, 61(1), 1–23. DOI: 10.1103/RevModPhys.61.1.

[5] Riess, A. G. et al. (2022). A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team. *The Astrophysical Journal Letters*, 934(1), L7. DOI: 10.3847/2041-8213/ac5c5b.

[6] DESI Collaboration (2025). DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations. *Journal of Cosmology and Astroparticle Physics*, 2025(02), 021. DOI: 10.1088/1475-7516/2025/02/021.

[7] Pfeifer, C. et al. (2025). From kinetic gases to an exponentially expanding universe – the Finsler-Friedmann equation. *Journal of Cosmology and Astroparticle Physics*, 2025(10), 050. DOI: 10.1088/1475-7516/2025/10/050.

[8] Trivedi, O. & Venkatasubramanian, V. (2026). Game theory in cosmology. *Physics of the Dark Universe*, 52, 102277. DOI: 10.1016/j.dark.2026.102277.

[9] Caldwell, R. R., Dave, R. & Steinhardt, P. J. (1998). Cosmological Imprint of an Energy Component with General Equation of State. *Physical Review Letters*, 80(8), 1582–1585. DOI: 10.1103/PhysRevLett.80.1582.

[10] Starobinsky, A. A. (1980). A New Type of Isotropic Cosmological Models Without Singularity. *Physics Letters B*, 91(1), 99–102. DOI: 10.1016/0370-2693(80)90670-X.

[11] Sotiriou, T. P. & Faraoni, V. (2010). $f(R)$ Theories of Gravity. *Reviews of Modern Physics*, 82(1), 451–497. DOI: 10.1103/RevModPhys.82.451.

[12] Verlinde, E. (2011). On the Origin of Gravity and the Laws of Newton. *Journal of High Energy Physics*, 2011, 29. DOI: 10.1007/JHEP04(2011)029.

[13] Verlinde, E. (2017). Emergent Gravity and the Dark Universe. *SciPost Physics*, 2(3), 016. DOI: 10.21468/SciPostPhys.2.3.016.

[14] Padmanabhan, T. (2012). Emergent Perspective of Gravity and Dark Energy. *Research in Astronomy and Astrophysics*, 12(8), 891. DOI: 10.1088/1674-4527/12/8/003.

[15] Euclid Collaboration (2025). Euclid Quick Data Release (Q1) – Data release overview. ESA/Euclid Consortium. DOI: 10.48550/arXiv.2503.15302.

[16] Casimir, H. B. G. (1948). On the attraction between two perfectly conducting plates. *Proceedings of the Royal Netherlands Academy of Arts and Sciences*, 51, 793–795.

[17] Hawking, S. W. (1974). Black hole explosions? *Nature*, 248, 30–31. DOI: 10.1038/248030a0.

[18] Nash, J. F. (1950). Equilibrium points in n-person games. *Proceedings of the National Academy of Sciences*, 36(1), 48–49. DOI: 10.1073/pnas.36.1.48.

[19] DESI Collaboration (2025). DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints. *Physical Review D*, 112, 083515. DOI: 10.1103/tr6y-kpc6.

[20] Scolnic, D. et al. (2022). The Pantheon+ Analysis: The Full Data Set and Light-curve Release. *The Astrophysical Journal*, 938(2), 113. DOI: 10.3847/1538-4357/ac8b7a.

[21] Kass, R. E. & Raftery, A. E. (1995). Bayes Factors. *Journal of the American Statistical Association*, 90(430), 773–795. DOI: 10.1080/01621459.1995.10476572.

[22] Foreman-Mackey, D. et al. (2013). emcee: The MCMC Hammer. *Publications of the Astronomical Society of the Pacific*, 125(925), 306–312. DOI: 10.1086/670067.

[23] Labbé, I. et al. (2023). A population of red candidate massive galaxies ∼600 Myr after the Big Bang. *Nature*, 616(7956), 266–269. DOI: 10.1038/s41586-023-05786-2.

[24] Boylan-Kolchin, M. (2023). Stress testing ΛCDM with high-redshift galaxy candidates. *Nature Astronomy*, 7, 731–735. DOI: 10.1038/s41550-023-01937-7.

[25] Asencio, E., Banik, I. & Kroupa, P. (2023). The El Gordo galaxy cluster challenges ΛCDM for any plausible collision velocity. *The Astrophysical Journal*, 954(2), 162. DOI: 10.3847/1538-4357/ace62a.

[26] Miller, T. B. et al. (2018). A massive core for a cluster of galaxies at a redshift of 4.3. *Nature*, 556(7702), 469–472. DOI: 10.1038/s41586-018-0025-2.

[27] Milgrom, M. (1983). A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *The Astrophysical Journal*, 270, 365–370. DOI: 10.1086/161130.

[28] Skordis, C. & Złośnik, T. (2021). New Relativistic Theory for Modified Newtonian Dynamics. *Physical Review Letters*, 127(16), 161302. DOI: 10.1103/PhysRevLett.127.161302.

[29] Bekenstein, J. D. (2004). Relativistic gravitation theory for the modified Newtonian dynamics paradigm. *Physical Review D*, 70(8), 083509. DOI: 10.1103/Phys-

RevD.70.083509.

[30] Hu, W. & Sawicki, I. (2007). Models of $f(R)$ Cosmic Acceleration that Evade Solar-System Tests. *Physical Review D*, 76(6), 064004. DOI: 10.1103/PhysRevD.76.064004.

[31] Jacobson, T. (1995). Thermodynamics of Spacetime: The Einstein Equation of State. *Physical Review Letters*, 75(7), 1260–1263. DOI: 10.1103/PhysRevLett.75.1260.

[32] Martinez, J. N., Rodriguez-Ruiz, J. F. & Rodriguez, Y. (2026). Jacobson's thermodynamic approach to classical gravity applied to non-Riemannian geometries: remarks on the simplicity of Nature. *Preprint*, arXiv:2602.00422. DOI: 10.48550/arXiv.2602.00422.

[33] Dewar, R. (2003). Information theory explanation of the fluctuation theorem, maximum entropy production and self-organized criticality in non-equilibrium stationary states. *Journal of Physics A: Mathematical and General*, 36(3), 631–641. DOI: 10.1088/0305-4470/36/3/303.

[34] Wolpert, D. H. (2006). Information Theory – The Bridge Connecting Bounded Rational Game Theory and Statistical Physics. In: *Complex Engineered Systems*, Springer, 262–290. DOI: 10.1007/3-540-32834-3_12.

[35] Harris, C. R. et al. (2020). Array programming with NumPy. *Nature*, 585, 357–362. DOI: 10.1038/s41586-020-2649-2.

[36] Virtanen, P. et al. (2020). SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17, 261–272. DOI: 10.1038/s41592-019-0686-2.

[37] Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. *Computing in Science & Engineering*, 9(3), 90–95. DOI: 10.1109/MCSE.2007.55.

[38] Gelman, A. & Rubin, D. B. (1992). Inference from Iterative Simulation Using Multiple Sequences. *Statistical Science*, 7(4), 457–472. DOI: 10.1214/ss/1177011136.

[39] Chevallier, M. & Polarski, D. (2001). Accelerating Universes with Scaling Dark Matter. *Int. J. Mod. Phys. D*, 10, 213. DOI: 10.1142/S0218271801000822.

[40] Linder, E. V. (2003). Exploring the Expansion History of the Universe. *Phys. Rev. Lett.*, 90, 091301. DOI: 10.1103/PhysRevLett.90.091301.

[41] Blas, D., Lesgourgues, J. & Tram, T. (2011). The Cosmic Linear Anisotropy Solving System (CLASS) II: Approximation schemes. *Journal of Cosmology and Astroparticle Physics*, 2011(07), 034. DOI: 10.1088/1475-7516/2011/07/034.

[42] Geiger, L. (2026). Microscopic Foundations of the Curvature Relaxation Model: From Quantum Geometry to Macroscopic Saturation. Companion paper (Paper III), submitted simultaneously. DOI: 10.5281/zenodo.18728936.

[43] Geiger, L. (2026). The Galactic–Cosmological Nexus: Connecting CRM to MOND Dynamics via Curvature Saturation. Companion paper (Paper IV), submitted simultaneously. DOI: 10.5281/zenodo.18728936.

[44] G. Kovacs (2026). Gravity Imprints, Time Fades: A Curvature-Memory Relaxation Model with Exact MOND Steady State. Non-peer-reviewed manuscript v4.4, March 2026. Available at `https://vixra.org/abs/2604.0006`.